

Florian Handke

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Über mich

I combine engineering pragmatism with technical expertise in applied cryptography and industrial cybersecurity. With a background in mechanical engineering, I understand how the industry works. I then moved into software and security, gaining in-depth knowledge through hands-on experience.

I thrive on taking ownership, leading technical teams, speaking at conferences, teaching students and engineers, and building solutions that solve real-world problems. I love details, thinking end-to-end, and presenting complex content to different audiences. I specialize in turning complex security concepts into practical engineering solutions that balance technical feasibility with cost-effectiveness—because security is always a tradeoff.

Kernkompetenzen

Applied Cryptography

- Public Key Infrastructure (PKI)
- Certificate Lifecycle Management
- Device Identity & Secure Device Identity
- Secure Boot & Secure Firmware Updates
- Code Signing
- Hardware Security Modules (HSMs)
- PKCS#11 API Implementation (Python)
- PKCS#x Standards
- EST, CMP, OCSP Protocols
- ASN.1 (the challenging kind!)
- Keyfactor EJBCA & SignServer

IoT & Connectivity

- OPC UA (Implemented GDS Push)
- MQTT Protocol
- Grafana & Visualization
- Telegraf Data Collection
- InfluxDB Time-Series Database
- Node-RED (Published 2 NPM packages)
- Mosquitto MQTT Broker
- Mender OTA Updates
- Raspberry Pi & ESP32 Development

Industrial Security

- IEC 62443 Compliance
- Cyber Resilience Act (CRA)
- NIS2 Directive
- Security Architecture & Risk Assessment
- Industrial IoT & OT Security
- Expertise in IEEE 802.1 AR device identities (IDeVID & LDeVID)
- Expertise in zero-touch onboarding (BRSKI, FDO)

Software Engineering

- Python (Django, uv)
- Shell Scripting
- Docker & Containerization
- GitHub Actions & CI/CD
- CodeQL & Static Code Analysis
- SBOM (CycloneDX, SPDX)
- OpenSSF Best Practices & Badges
- OWASP ZAP Security Testing
- Static Typing & Linting
- Test Coverage & Automated Testing
- Dependency Management

Berufserfahrung

Sept. 2023 – Present · 2 years 11 months

Head of Industrial Security

Campus Schwarzwald

Jan. 2021 – Aug. 2023 · 2 years 8 months

Smart Production Engineer

Campus Schwarzwald

Apr. 2018 – Dec. 2020 · 2 years 9 months

Data Scientist: Condition Monitoring of Laser Devices

TRUMPF

Oct. 2015 – Sept. 2016 · 1 year

Customer and Project Manager

Andreas Lupold Hydrotechnik GmbH

Ausbildung

2016 – 2018

Master of Science (M.Sc.), Advanced Precision Engineering

Hochschule Furtwangen

2012 – 2015

Bachelor of Engineering (B.Eng.), Production Engineering

Duale Hochschule Baden-Württemberg

2011 – 2012

University of Applied Sciences Entrance Qualification

Berufskolleg Horb a.N.

2007 – 2011

Apprenticeship as a Precision Mechanic

Koch Pac Systeme GmbH

Ausgewähltes Projekt

Trustpoint – Open Source Trust Anchor

Co-created and technically led the development of Trustpoint, an open-source trust anchor and PKI platform developed together with contributors from multiple organizations. Trustpoint is a platform for managing digital machine identities and certificate lifecycles in industrial environments. It supports secure device onboarding, certificate issuance, renewal, revocation, and decommissioning using standardized protocols and can operate locally in segmented, offline, or air-gapped OT networks. By combining automation, user-friendly workflows, and vendor-independent standards, Trustpoint helps manufacturers, integrators, and operators reduce security risks and implement trustworthy industrial communication without requiring deep PKI expertise.

GitHub: <https://github.com/Trustpoint-Project/trustpoint>

Vorträge & Veröffentlichungen (Auswahl)

**Automated Identity Management for
Embedded Components**

embedded world conference

March 2026

Industrial trust anchor

all about automation

March 2026

Digital Identities for the industry

SPS Messe

November 2025

Industrial Security at Campus Schwarzwald

OT Security Breakfast

Oktober 2025

**Cryptography and Legal Requirements for
Businesses**

Campus Schwarzwald

Mai 2025

**Certificate Management Made Easy –
Practical Solutions for Machinery and Plant
Operators**

Hannover Messe

April 2025

**Securing Industry 4.0 with Digital Identities
for Embedded Components**

embedded world conference

March 2025

**The rising importance of managed
certificates in industrial systems**

embedded world conference

March 2025

**From Emmental to Gouda: Why
Cybersecurity Is Now a Management Issue**

Business Breakfast

February 2025

Digital trust

Campus Cyber Paris

December 2024

Lehre & Community

Teaching & Public Speaking

- Part-time lecturer at DHBW Stuttgart (Industrial Security)
- Part-time lecturer at University of Stuttgart (Industrial Security)
- Industrial security trainings & workshops ( Applied Cryptography Training)
- Conference speaker & presenter
- Author of technical articles (ATP, embedded world conference, IT-SICHERHEIT, IT&Produktion)

Community & Events

- VDMA Industrial Security Working Group
- Open Industry 4.0 Working Group
- Organizer: PKI in der Automatisierungstechnik (since 2023)
- Organizer: OT-Security Forum
- Organizer: OT-Security Breakfast

Stand: Juli 2026